Strava Fitness Data Analysis Project Report

1. Project Overview

The Strava Fitness Data Analysis Project aims to analyze user fitness activity data to uncover patterns related to physical activity, sleep behavior, and health metrics such as weight and BMI.

In this project, data from multiple datasets was cleaned, analyzed, and visualized using Python. The processed data was then stored and analyzed using SQL, followed by interactive dashboard creation using Power BI to present insights in a visual format.

2. Problem Statement

Fitness applications collect large amounts of user health data, but without proper analysis it is difficult to understand user behavior and extract meaningful insights.

The challenge is to analyze and integrate different datasets related to activity, sleep, and weight in order to identify patterns that can help improve fitness awareness and support better health decisions.

3. Business Objectives

- Analyze daily physical activity patterns of users.
- Evaluate sleep duration and sleep quality.
- Study weight and BMI trends among users.
- Identify peak activity hours from hourly step data.
- Examine relationships between activity levels, sleep patterns, and health metrics.

4. Tools and Technologies Used

Python – Data Cleaning, EDA, Data Transformation

Pandas – Data manipulation

Matplotlib / Seaborn – Data visualization

MySQL – Data storage and querying

Power BI – Dashboard creation and visualization

5. Dataset Description

The project uses four datasets containing different types of user fitness data:

1. Daily Activity Dataset – Steps, calories burned, active minutes.

2. Sleep Dataset – Total minutes asleep, total time in bed.

3. Weight Log Dataset – Weight and BMI.

4. Hourly Steps Dataset – Steps recorded every hour.

These datasets were later merged to perform deeper analysis and identify relationships between activity and health metrics.

6. Data Cleaning (Python)

Data cleaning was performed using Python to ensure data quality before analysis.

Steps performed:

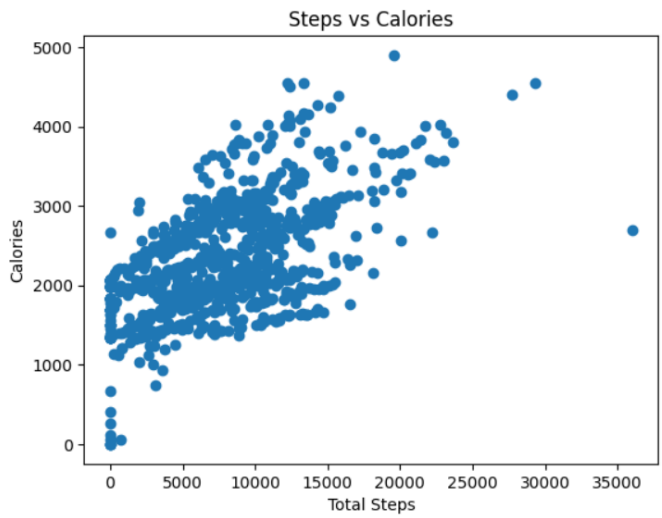
- Handling missing values
- Removing duplicate records
- Converting date columns into datetime format
- Checking dataset structure
- Standardizing column formats
- Preparing datasets for merging

7. Exploratory Data Analysis (EDA)

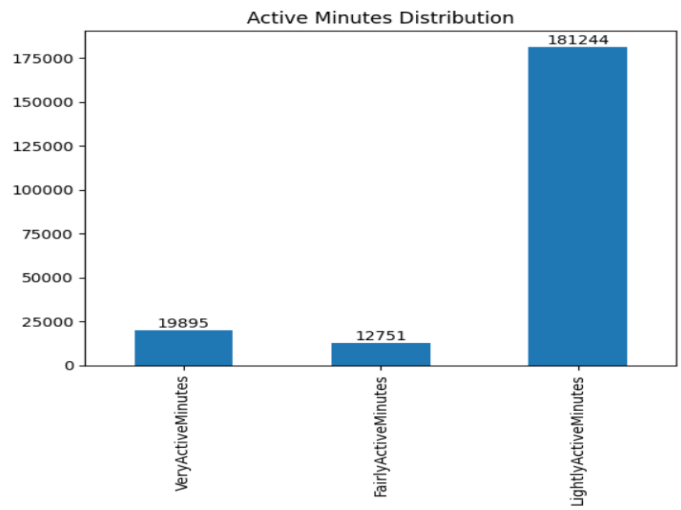
EDA was conducted to understand trends and patterns in the dataset.

A total of 19 visualizations were created including:

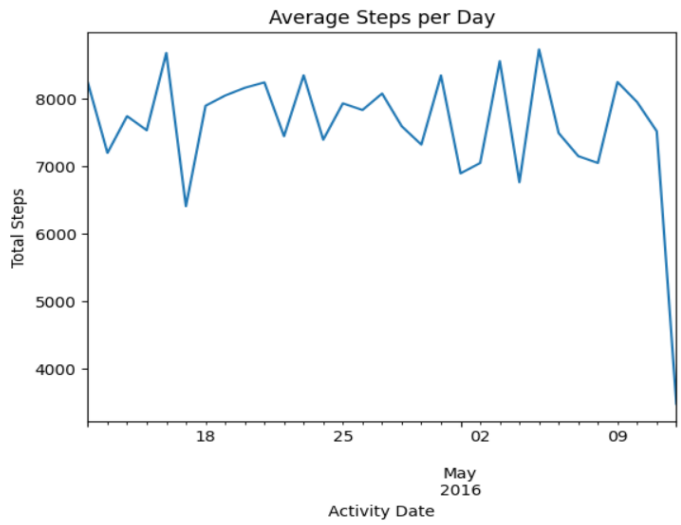
• Steps vs Calories relationship



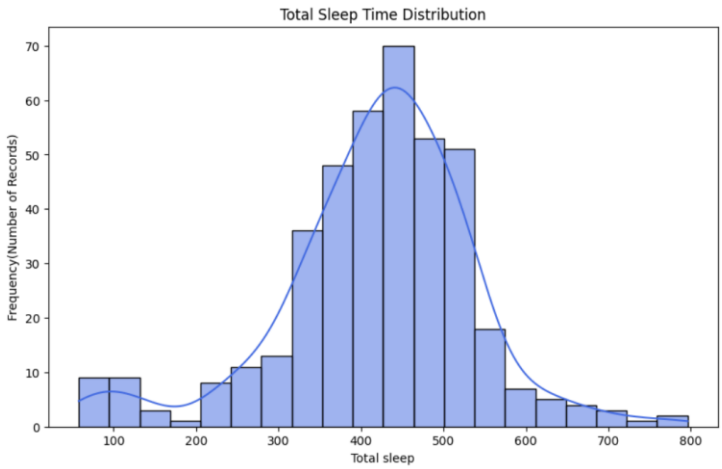
• Activity intensity distribution



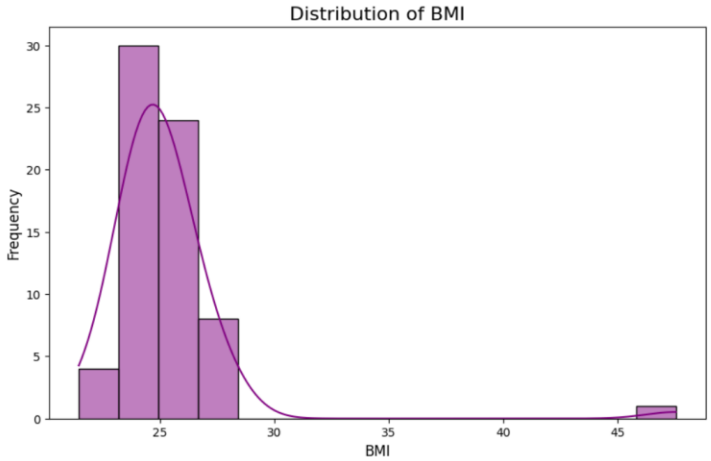
• Daily step trends



• Sleep duration patterns



• Distribution of BMI



8. SQL Database Implementation

After cleaning and merging the data, datasets were imported into MySQL.

Steps performed:

• Database creation

• Importing cleaned datasets

• Creating relationships between tables

• Executing analytical SQL queries

Queries included:

• Average steps per user

Id	avg_daily_steps
8877689391	16040.0323
8053475328	14763.2903
1503960366	12116.7419
2022484408	11370.6452
7007744171	11323.4231
3977333714	10984.5667
4388161847	10813.9355
6962181067	9794.8065

• Top active users

Id	Total_steps
8877689391	497241
8053475328	457662
1503960366	375619
2022484408	352490
4388161847	335232
3977333714	329537
6962181067	303639
7007744171	294409

• Sleep duration trends

Id	avg_sleep_minutes
1844505072	652.0000
2026352035	506.1786
6117666160	478.7778
4319703577	476.6538
5553957443	463.4839
7086361926	453.1250
6962181067	448.0000
2347167796	446.8000

• Activity vs Weight

Id	avgSteps	avgWeight
1503960366	12116.7419	52.5999984741211
1927972279	916.1290	133.5
2873212765	7555.7742	57.00000000000001
4319703577	7268.8387	72.35000228881836
4558609924	7685.1290	69.63999938964844
5577150313	8304.4333	90.6999969482422
6962181067	9794.8065	61.553333791097
8877689391	16040.0323	85.14583428700765

• Peak activity hours

hourofday	avgSteps
18	599.1700
19	583.3907
17	550.2329
12	548.6421
14	540.5136
13	537.6982
16	496.8456
10	481.6652

9. Power BI Dashboard Development

Interactive dashboards were created in Power BI to visualize insights.

Dashboard 1 – Activity Overview

• Total steps

• Calories burned

• Activity intensity

• Daily step trends

• Activity level distribution

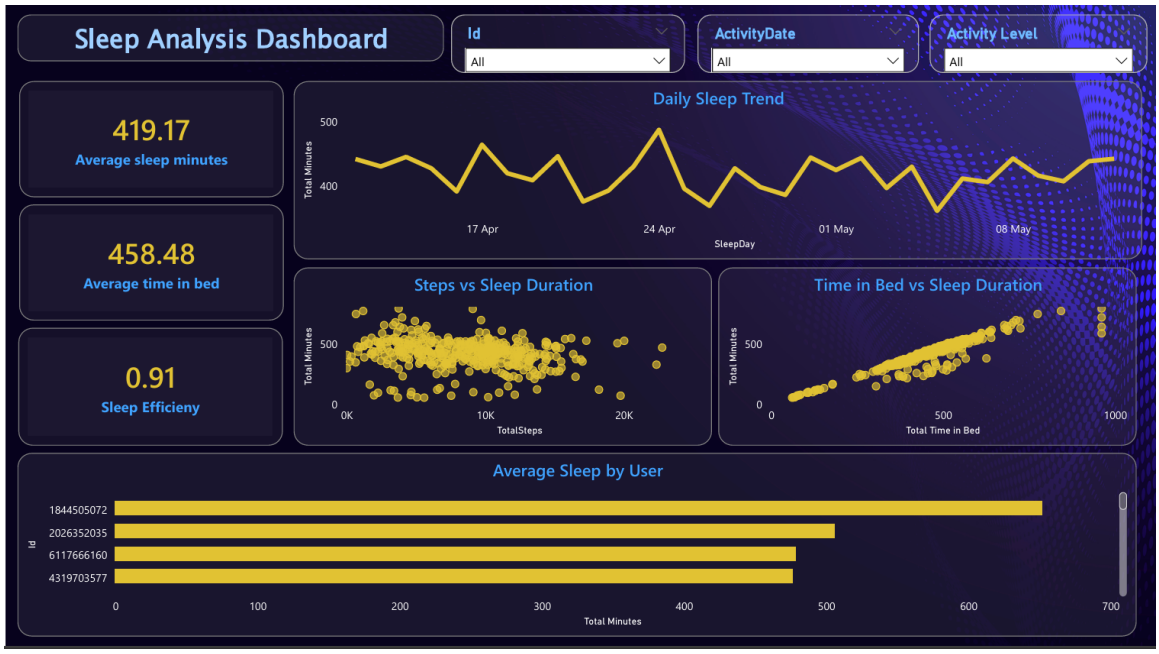
Dashboard 2 – Sleep Analysis

• Average sleep duration

• Sleep efficiency

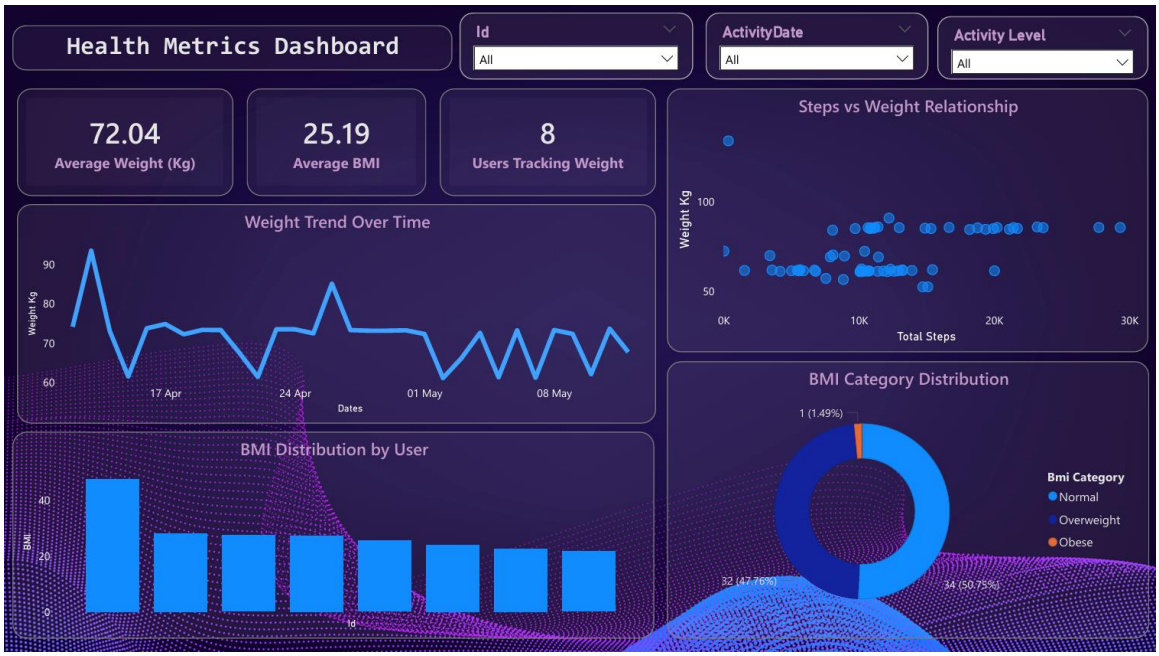
• Sleep trends

• Steps vs sleep relationship



Dashboard 3 – Health Metrics

- Average weight
- Average BMI
- BMI category distribution
- Weight trend over time
- Steps vs weight relationship



10. Key Insights

• Higher step counts generally result in higher calorie burn.

• Sleep duration across users remains relatively stable.

• Users show high sleep efficiency (~90%).

• Most users fall in normal to overweight BMI categories.

• Weak correlation observed between steps and weight.

• Activity levels vary among users.

11. Conclusion

This project demonstrates how fitness data can be analyzed to generate meaningful insights into user health and activity behavior.

By combining Python for data processing, SQL for querying, and Power BI for visualization, the project presents a complete analytics workflow from raw data to actionable insights.

Such analysis helps understand how physical activity impacts health metrics and supports better health monitoring systems.